

Michael J. Sieler Jr., PhD | Data Scientist | Computational Biology | Multi-Omics Analytics
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Summary

Data scientist with 8+ years of experience developing and applying computational, statistical, and experimental approaches to complex biological datasets. Expertise in multivariate modeling, longitudinal data analysis, multi-omic integration, reproducible workflows, and FAIR data practices. Proven collaborator across international, interdisciplinary, and cross-functional teams, with a strong record of translating complex data into actionable scientific insights through publications, workshops, and collaborative research support.

Skills

R, Python, Bash/Unix, Git • Statistical modeling • Multivariate analysis • Longitudinal data analysis • Data visualization • Multi-omic integration • Microbiome, transcriptomic, and metabolomic data • Reproducible workflows • FAIR data practices • Cross-functional collaboration • Scientific communication • Training and mentorship

Experience

Postdoctoral Researcher, Computational Biology | University of Geneva, Switzerland | 2026 to Present

- Lead analysis of complex multi-source biological datasets to support research on host-microbiota interactions and behavior.
- Build reproducible analytical workflows in R and Python for data cleaning, integration, visualization, and interpretation across collaborative projects.
- Partner with international and interdisciplinary collaborators to align analytical strategy with scientific goals and communicate findings across teams.

Postdoctoral Researcher, Microbiome Data Science | Oregon State University, USA | 2025

- Applied statistical modeling and data visualization approaches to analyze complex microbiome datasets and support interpretation across collaborative research projects.
- Contributed quantitative expertise to study design, analysis planning, and communication of results for host-associated microbiome research.

Expert Biology AI Data Labeler | HandshakeAI, USA | 2025

- Contributed biological domain expertise to development of 100+ LLM prompts and evaluation tasks for scientific AI applications.
- Improved biological accuracy, consistency, and scientific reasoning in AI-generated outputs through expert review and feedback.

PhD Researcher, Microbiome and Multi-Omics Data Science | Oregon State University, USA | 2020 to 2025

- Designed and executed longitudinal, multifactorial studies integrating microbiome, host transcriptomic, and physiological data across large experimental cohorts.
- Built computational and statistical workflows for data cleaning, normalization, modeling, and visualization of complex multi-omic datasets.
- Led quantitative analyses examining how environmental and biological perturbations affect host-associated microbial systems.
- Communicated findings through first-author publications, invited talks, workshops, and cross-laboratory collaborations.

FAIR Microbiome Data Fellow | National Microbiome Data Collaborative, USA | 2024

- Supported development and dissemination of microbiome data workflows aligned with FAIR data principles and metadata standards.
- Trained researchers in best practices for data organization, interoperability, and reproducible microbiome research workflows.

PhD Bioinformatics and Data Science Intern | Pacific Northwest National Laboratory, USA | 2023

- Developed a reproducible statistical pipeline in R to analyze hundreds of lipidomic samples from controlled experimental datasets.
- Evaluated batch-effect correction approaches and contributed to collaborative bioinformatics method development.

Education

PhD, Microbiology | Oregon State University | 2025

BS, Bioresource Research, Bioinformatics and Genomics | Oregon State University | 2020